MATH 152 Fall 2019 Section 06---Prof. Dean

MIDTERM EXAM 3

Tues. December 3, 2019

NAME (please print legibly): _____

Student ID Number: _____

Discussion Section (please circle): 07: Gerson (9:00) 08: Abhishek (9:00)

09: Karan (10:00) 10: Abhishek (10:00)

- No calculators or notes are allowed on this exam.
- Please show all your work. You may use the backs of pages if necessary. You may not receive full credit for a correct answer if there is no work shown.
- Don't cheat.

QUESTION	VALUE	SCORE
1	20	
2	20	
3	20	
4	10	
5	20	
6	20	
TOTAL	100	

1) **(20 points) Find the radius of convergence and interval of convergence of the following power series.**

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n (x-3)^n}{n^{2/3}\, 4^n}$

(b) $\sum_{n=0}^{\infty} \frac{(2n)!\,(x-2)^n}{n!}$

2) (20 points) (a) Find a power series representation for the function

$$f(x) = \frac{1}{9-x^2}$$

What is the radius of convergence of this series?

(b) Use the series you found in part (a) to find a power series representation for the function

$$g(x) = \frac{x^3}{(9-x^2)^2}$$

What is the radius of convergence of this series?

3) (20 points) (a) Use the definition of Taylor series to find the Taylor series of $f(x) = \frac{1}{2x+7}$ centered at $a = 1$. (You do not need to find the radius of convergence.)

(b) Evaluate the following integral as an infinite series. (You do not need to find the radius of convergence.)

$$\int x^2 \sin(3x^2)\, dx$$

4) **(10 points) Let** $f(x) = (3x - 1)^{5/3}$

(a) **Approximate** f **by a Taylor polynomial with degree 2 centered at** $a = 3$.

(b) **Use Taylor's Inequality to estimate the accuracy of the approximation** $f(x) \approx T_2(x)$ **when** x **lies in the interval** $2.95 \leq x \leq 3.05$. **(You do not need to simplify your answer.)**

5) (20 points) (a) Find the center, foci, and vertices of the ellipse and sketch the graph.

$$\boldsymbol{4x^2 + y^2 + 8x - 4y = 8}$$

(b) Find an equation of the parabola with focus $(3, 4)$ and directrix $y = -2$. Sketch its graph.

6) **(20 points) (a) Sketch the parametric curve and eliminate the parameter to find the Cartesian equation of the curve. (Simplify your answer.)**

$$x = 4 - t,\ \ y = t^2 - 2t,\ \ 0 \le t \le 4$$

(b) Find the arc length of the curve.

$$x = 4 - 3t^2,\ \ y = 2t^3 + 1,\ \ 0 \le t \le 4$$